DINITHI DISSANAYAKE

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Interdisciplinary PhD researcher in Computer Vision and Human-Computer Interaction, focused on multimodal AI for wearable and egocentric systems. I develop machine learning and deep learning models that combine first-person video, audio, gaze, and physiological signals to infer user context and support adaptive assistance. Experienced in multimodal data pipelines, user studies, and real-time wearable/Virtual Reality prototyping.

EDUCATION

PhD in Computer Science *Aug 2023 – Present*
School of Computing, National University of Singapore, Singapore
Supervised by Prof. Suranga Nanayakkara

Bachelor of Science in Engineering, Electronic and Telecommunication Engineering *Aug 2017 – Jul 2022*
University of Moratuwa, Sri Lanka
First Class Honors | GPA: 3.92/4.20 | Dean's List: 7/8 semesters

TECHNICAL SKILLS

Programming: Python, C++, SQL, Git, Matlab
ML, Vision & Development Tools: PyTorch, TensorFlow, OpenCV, Unity, Docker, AWS, GCP
Wearables & Sensing: Meta Aria Glasses, Meta Quest XR Headsets, Tobii Eye Tracker, Shimmer Sensors, Garmin Sensors, EDA, PPG, IMU, HRV
Computer Vision & Machine Learning Focus Areas: Egocentric video understanding, gaze estimation, activity recognition, 3D point cloud understanding, multimodal fusion, affective and behavior analysis, temporal modeling

EXPERIENCE

Research Intern *Aug 2024 – Feb 2025*
Meta Reality Labs × Augmented Human Lab, NUS, Singapore

- Designed and executed a large-scale VR gameplay study (N=150) using eye-tracking, head-motion, and physiological signals to model motion sickness onset.
- Developed machine learning models to predict real-time discomfort scores during VR gameplay on the Meta Horizon Worlds platform.
- Built robust pipelines for synchronizing high-frequency HMD, physiological, and eye-tracking data streams, including data processing and feature engineering.

Data Analytics Consultant *Jan 2023 – Aug 2023*
LIRNEasia, Colombo, Sri Lanka

- Developed machine learning models and data pipelines to classify built-up regions in satellite imagery, supporting a data-driven urbanization index for Sri Lanka.
- Integrated QGIS and Google Earth Engine workflows for automated spatial analysis and visualization, and time series analysis over built-up areas.
- Designed longitudinal datasets to analyze household energy consumption patterns.

Data Engineer *May 2022 – Aug 2023*
Axiata Digital Labs Pvt. Ltd., Colombo, Sri Lanka

- Engineered ML/DL deployment pipelines for the company's AI Factory platform, supporting model development, testing, deployment, and integration into business workflows.
- Built scalable data ingestion and model integration workflows for computer vision, time-series forecasting/regression, and customer analytics use cases using cloud-based infrastructure.
- Developed a customer churn prediction model and deployment pipeline for a major telecom client, enabling data-driven customer retention analytics.

SELECTED PUBLICATIONS

HeadRoom: Lightweight, Edge-deployable Pipeline for Adaptive Notification Routing. Conditionally Accepted at *UIST '26*.
Dinithi Dissanayake, Prasanth Sasikumar, Suranga Nanayakkara

Two-Stage Multimodal Framework for Emotion Mimicry Intensity Prediction. *CVPR Workshops '26*.
Dinithi Dissanayake, Shaveen Silva, Ovindu Atukorala, Prasanth Sasikumar, Suranga Nanayakkara

Interrupting Autopilot: Evaluating Sensory Interventions for Momentary Self-Regulation. *CHI EA '26*.
Dinithi Dissanayake, Muftee Mysan, Yifei Luo, Timothy Antoni, Peter Cleveland, Prasanth Sasikumar, Suranga Nanayakkara

VRSense: An Explainable System to Help Mitigate Cybersickness in VR Games. *CHI EA '25*.
Dinithi Dissanayake, Chitralekha Gupta, Prasanth Sasikumar, Suranga Nanayakkara

Navigating the State of Cognitive Flow: Context-Aware AI Interventions for Effective Reasoning Support. *CHI '25 Workshop on Human-AI Interaction for Augmented Reasoning*.
Dinithi Dissanayake, Suranga Nanayakkara

CrossPoint: Self-Supervised Cross-Modal Contrastive Learning for 3D Point Cloud Understanding. *CVPR '22*.
Aflal Afham, Isuru Dissanayake, **Dinithi Dissanayake**, Amaya Dharmasiri, Kanchana Thilakarathna, Ranga Rodrigo

3DLatNav: Navigating Generative Latent Spaces for Semantic-Aware 3D Object Manipulation. *ECCV Workshops '22*.
Amaya Dharmasiri, **Dinithi Dissanayake**, Isuru Dissanayake, Aflal Afham, Ranga Rodrigo, Kanchana Thilakarathna

A Systematic Literature Review of User-Aware Adaptive Assistive Wearables. Under review at *ACM Computing Surveys*.
Dinithi Dissanayake, Hyong Woon Lee, Moritz Messerschmidt, Yize Wei, Suranga Nanayakkara

SELECTED RESEARCH PROJECTS

Doctoral Thesis: Multimodal AI for Context-Aware Wearable Assistance *Ongoing*

- Developing multimodal models that combine first-person video, audio, gaze, and physiological signals using transformer-based encoders and MViTs for user-context inference and human behavior modeling (gaze, activity).
- Drawing on neuroscience-inspired principles of sensory processing and attention to model sensory load and cognitive demand in wearable and egocentric settings.
- Designing adaptive fusion and assistance strategies that dynamically prioritize informative modalities and support timely intervention under changing user and environmental conditions.
- Optimizing models into efficient wearable-deployable pipelines through model simplification, runtime optimization, and deployment-aware evaluation.
- Designing user studies to quantify the impact of adaptive assistance on reaction time, task performance, and user experience.

Motion Sickness Modeling and Mitigation in Virtual Reality Gameplay *Oct 2023 – Feb 2025*

- Designed and conducted a large-scale VR gameplay study (N=150) using physiological signals, HMD data, gaze behavior, and self-reported discomfort.
- Built multimodal pipelines to synchronize physiological, eye-tracking, and headset sensor streams for cybersickness prediction.
- Developed pipelines for real-time ML and explainable AI modules to detect cybersickness onset and identify key contributing features.
- Interviewed expert VR developers to translate model explanations into design insights for reducing cybersickness in VR games.

3D Object Transformation and Regeneration for Privacy in Mixed Reality *Apr 2022*

- Developed a 3D vision pipeline for object-part transformation, including adding, removing, and modifying object components in mixed-reality scenes.
- Designed 3D–2D correspondence methods to support point-cloud understanding and semantic-aware object manipulation.
- Evaluated the system against simulated privacy attacks and explored deployment feasibility on smart-device platforms.